

⑤ Disaster Recovery and Business Continuity Strategy for Cloud.

A. A cloud-based big data platform should have a disaster Recovery (DR) and Business Continuity (BC) strategy to ensure that data processing and business operations continue even when failure occur.

Strategy:

1. Redundancy: Deploy critical servers, storage systems, and services across multiple availability zones or regions. If one component fails, another can continue operating.
2. Data Replications:- Maintain multiple copies of important data in different locations. Synchronous replication can be used for critical data, while asynchronous replication can be used for large datasets.
3. Regular Backups: Perform automated backups of data bases, metadata, configurations and important data sets. Backup should be stored in a separate region.

4. Automated failover: Use load balancers and cloud orchestration systems to automatically redirect workloads to healthy servers when a failure is detected.
5. Monitoring and Alerting: Continuously monitor system health, storage, network, performance and application availability.
6. Recovery Objectives: Define RPO to determine acceptable data loss and RTO to determine acceptable recovery time.
7. Disaster Recovery Testing: Regularly test backup restoration and failover procedures to ensure that the recovery plan actually works.

Justification:-

Redundancy eliminates single points of failure, replication protects against data loss and automated failover reduces downtime by quickly switching to healthy resources. Together, these mechanisms provide high availability, fault tolerance, data protection, and operational resilience.

6. Impact Of Data Quality issues & Data Governance Strategies

Data quality is extremely important in a large-scale analytics platform because poor quality data can produce incorrect analytical results and business decisions.

Major data quality issues:

- Inconsistency: Different systems may store the same information in different values. This can produce conflicting analytical results.
- Duplications: Duplication records can cause incorrect counts, biased statistics, and inflated customer or transaction numbers.
- Missing values: Missing data can reduce the accuracy of prediction and may cause analytical algorithms to produce unreliable results.

For example, if a customer appears three times in a dataset, the analytics system may incorrectly conclude that there are 3 different customers.

Data governance strategies:-

1. Data validation: Apply rules to check data type, range, formats and mandatory fields.
2. Data cleansing: Remove duplicates, correct inconsistent values and handle missing values using suitable techniques.
3. Master Data management: maintain a single trusted version of important entities such as customers, products and locations.
4. Data Quality Monitoring: Continuously measure accuracy, completeness, consistency, and uniqueness.
5. Data Lineage: Track where data comes from, how it is transformed, and where it is used.
6. Access Control: Define permission so that only authorized users can modify important data sets.
7. Data Governance Policies: Establish standards for data ownership, storage, naming, security and quality.

Scalable Recommendation system for an Online Streaming Service.

A. A Streaming Service such as a movie or music platform can use big data technologies to build a Scalable Recommendation system.

Architecture :

1. Data Collection :

Collect user behaviour such as :

- Movies / songs watched or played
- Search history
- Ratings & likes
- Watch duration
- Skips
- Browsing history.

2. Distributed Data Storage :

Store large volumes of user and content data using distributed storage systems like HDFS or cloud object storage.

3. Data processing:- use distributed processors Frameworks such as Apache Spark to process millions of user interactions efficiently.

4. Feature Generation:- Generate features such as preferred genres, frequently watched content.

5. Recommendation Algorithms:-

— Collaborative filtering.

— Content based filtering

— Hybrid recommendation

6. Real Time Recommendation

streaming technologies can process recent user activity and update recommendations quickly.

7. Scalable Deployment:

Deploy the recommendation model using distributed cloud infrastructure so that additional computing resources can be added when the number of users increased.